Digitally Assisted DC-Offset Calibration of a Multistage Geophone Front End

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***Abstract*—Accumulated amplifier and reference offsets can saturate intermediate nodes in geophone front ends built around a programmable system-on-chip (PSoC), reducing the dynamic range available to microvolt-level seismic signals. This paper presents a foreground method that sequentially recenters each stage of the analog signal chain—from the programmable-gain amplifier (PGA) to the final low-pass filter (LP)—using the on-chip digital-to-analog and analog-to-digital converters (DAC, ADC). The dataset comprises ten stationary-geophone restarts under each of eight board/gain conditions on two independently assembled boards, each measured both calibrated and under a shared fixed-reference baseline. At the LP output delivered to the ADC, calibration reduced the sample-mean offset to 0.4–3.9% of the nominal 5-V span from 2.9–45.8%. The largest residual occurs at the PGA, consistent with its unfiltered wideband response. These descriptive results establish initial feasibility; confirmatory statistical validation and controlled field comparison remain future work.**

***Index Terms*—Analog front end, DC-offset calibration, geophone, mixed-signal systems, programmable system-on-chip (PSoC)**

I. Introduction

Multichannel surface-wave methods use spatially distributed receiver arrays to estimate Rayleigh-wave dispersion and infer shear-wave velocity profiles for geotechnical site characterization [1], [2]. Their processing is well established, but acquisition hardware remains a major source of cost and deployment complexity. Low-cost embedded geophone interfaces have been demonstrated [3], [4]; in such multistage chains, however, each amplifier and reference contributes an offset, and these accumulate, consuming converter input span and, at high gain, saturating intermediate nodes before the microvolt-level signal is digitized. Raising analog-component grade increases the cost of every node, and manual per-node trimming scales poorly: each receiver added to the array compounds the calibration effort. This work instead uses the configurable analog resources already present in a programmable system-on-chip (PSoC) to recenter a budget-constrained front end.

The proposed front end is part of a low-cost, reconfigurable Internet-of-Things (IoT) acquisition platform [5] based on SM-24 moving-coil geophones [6]. The PSoC 5LP integrates programmable analog blocks, voltage DACs (VDACs), a delta-sigma ADC, a digital filter, direct memory access (DMA), and an Arm Cortex-M3 processor [7]. Time-sharing the ADC and digital filter between acquisition and calibration avoids a separate converter or processor, following digitally assisted analog design [8]; the tradeoff is that the internal analog blocks cannot be independently selected to optimize noise and offset [9]. Only the analog calibration subsystem is evaluated here.

The conditioning circuit derives from the low-frequency geophone front end of Ma et al. [10]. Most stages remain DC-coupled to preserve the low-frequency transfer used for surface-wave dispersion analysis; coupling capacitors would insert high-pass poles, and placing their corners well below the band would require large, tolerance-sensitive off-chip capacitors. On the prototype, accumulated offsets and warm-up drift consumed operating range at intermediate and final nodes, so the method recenters them at each observable stage through a finite foreground calibration before acquisition (Section III), without altering the acquisition-path transfer function.

Prior offset correction follows three lines: numerical estimation and storage of digital corrections for a transimpedance amplifier, PGA, and ADC under internal stimulus [9], [11]; physical trimming of a single amplifier through body biasing [12]; and classical autozero or chopper techniques [13], which are unavailable in the selected PSoC blocks and would require external circuitry. In contrast, the proposed method physically adjusts the reference of each trim-accessible stage through its dedicated VDAC, reusing the existing ADC, digital filter, and multiplexer with no continuously active acquisition-time loop.

Sequential per-stage recentering followed by frozen VDAC codes reduces the accumulated offset delivered to the acquisition ADC without adding poles or requiring controller retuning. The contributions are: 1) a four-VDAC architecture reusing the PSoC multiplexer, ADC, and digital filter; 2) a bounded deadband controller that freezes its codes before acquisition; and 3) an exploratory two-board evaluation over eight board/gain conditions spanning $\times1$–$\times50$ with a single parameter set. Sections II and III present the architecture and controller, Section IV the experiments, and Sections V and VI the discussion and conclusions.

II. Platform Design

The PSoC conditions and digitizes the geophone output in acquisition mode. In calibration mode, the multiplexer, ADC, digital filter, and DAC-control resources are temporarily repurposed as a mixed-signal feedback loop, while the sensor signal chain remains unchanged.

In the Ma *et al.*-derived feed-forward signal chain of Fig. 1 [10], an offset introduced at any stage can propagate through downstream gains toward the ADC input. The path is not a strict one-input cascade: the band-pass filter (BP) follows the

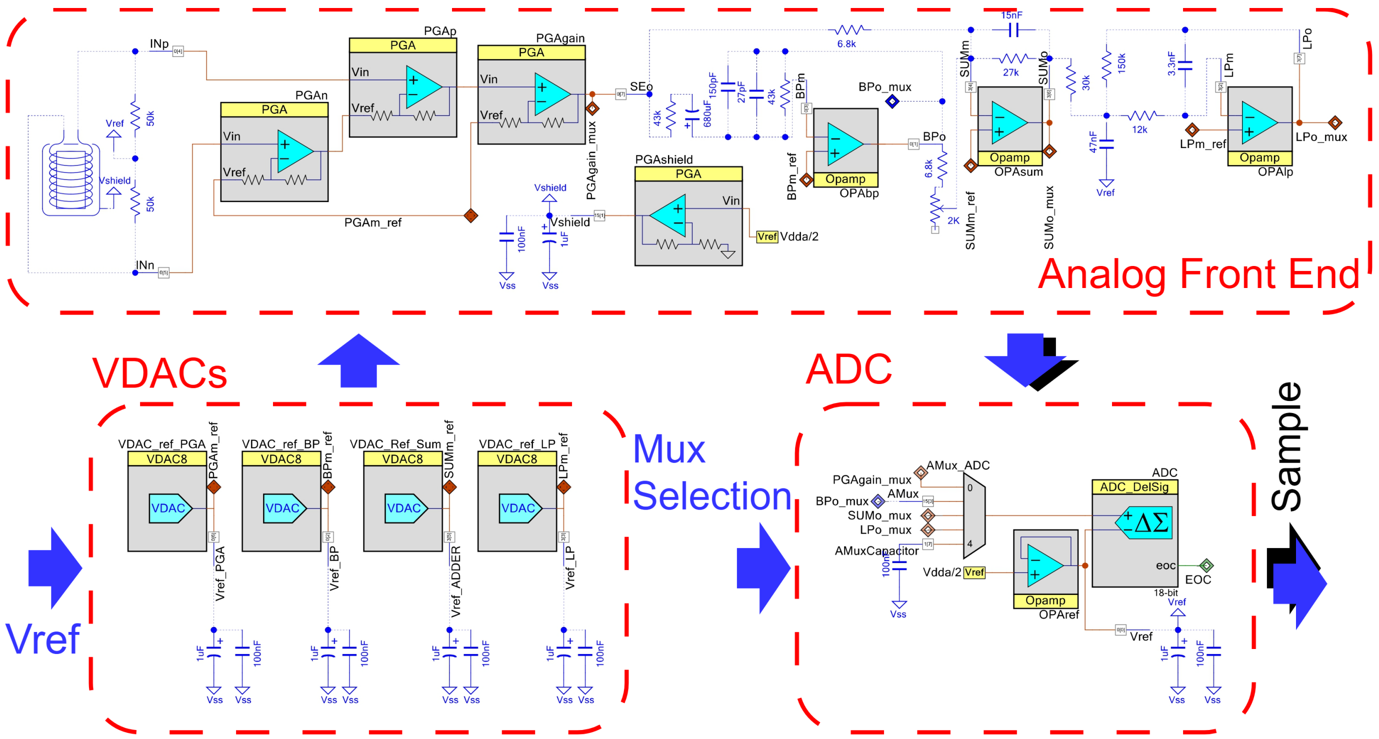

Fig. 1: Analog signal chain: four independent VDAC references trim the programmable-gain amplifier (PGA), band-pass filter (BP), summing stage (SUM), and low-pass filter (LP), with a shared ADC/multiplexer measurement path.

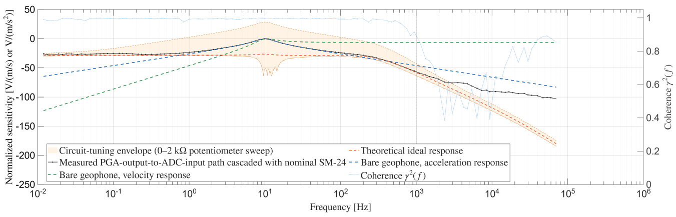

Fig. 2: Initial analog-path check: sinusoidal sweeps measured the PGA-output-to-ADC-input path, cascaded with the nominal SM-24 model (markers). Left: normalized magnitude, with the theoretical models and 0–2 kΩ tuning envelope as shape references. Right: local coherence. The dash-dotted line marks the rounded 1-kHz coherence onset. No calibrated seismic source available.

programmable-gain amplifier (PGA), whereas the summing stage (SUM) combines a direct PGA branch with the BP branch before the low-pass filter (LP). The calibration therefore exposes and trims each stage separately rather than correcting only the final ADC value.

An analog multiplexer connects one stage output at a time to the shared 18-bit delta-sigma ADC [7], the error sensor, with software mutual exclusion preventing simultaneous connections. A switched capacitor behind the selected MOSFET adds limited RC anti-aliasing with its on-resistance and isolates the amplifier from a direct capacitive load.

For the ith measured node, let y_i^{DC} denote its slowly varying voltage deviation from the intended stage reference, with polarity matching the differential ADC coordinates. It can be represented as

$$y_i^{\mathrm{DC}} = \sum_{j<i} A_{ij} y_j^{\mathrm{DC}} + b_i + K_i u_i, \tag{1}$$

where A_{ij} describes each actual DC feed from an upstream node j, b_i collects local amplifier, reference, resistor-mismatch, and component-tolerance contributions in voltage coordinates, u_i is the VDAC voltage, and K_i is the signed DC gain from that VDAC to the measured node. For the first stage (PGA), the upstream sum is empty. The firmware represents u_i by an 8-bit command code c_i, such that $u_i = q_{\mathrm{DAC}} c_i$. The multiple-input form covers both ordinary cascaded stages and nodes such as SUM; the calibration target is $y_i^{\mathrm{DC}} = 0$.

The architecture assigns one u_i to each trim-accessible node. With negligible reverse coupling, sensor and filter-state changes small relative to the selected deadband during DC estimation, and adequate VDAC authority and range, Eq. 1 motivates a forward pass that brings each measured y_i^{DC} toward zero in PGA, BP, SUM, and LP order (Section III). Equation 1 is an equilibrium model; it does not describe the finite settling of amplifier and analog/digital filter states after a stage or coefficient switch.

III. Self-Calibrating Geophone Platform

A. *Digital Hardware and Calibration Sequence*

At boot, after a gain change, or on command, the firmware loads stored DAC codes and verifies the offsets (Fig. 3). Because DC estimation assumes a stationary geophone and slow internal-state change over the lock window, an accepted

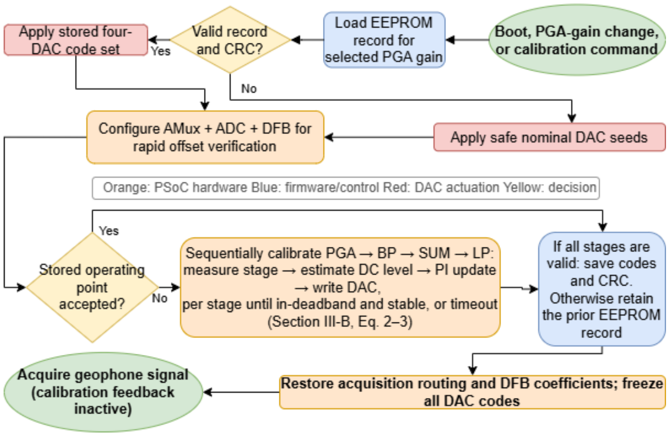

Fig. 3: Foreground calibration flow: stored-code verification, sequential trimming, and acquisition.

operating point returns directly to acquisition; otherwise PGA, BP, SUM, and LP are processed sequentially, with filter reset and settling after each switch. A watchdog and sample limits bound the sequence; during acquisition the controller is inactive, LP is selected, and the VDAC codes are constant.

The shared digital filter is reconfigured between modes: acquisition loads coefficients tailored to the geophone's operational bandwidth, whereas calibration uses a 128-tap low-pass finite-impulse-response (FIR) to estimate the DC offset, with direct memory access moving data through RAM. After calibration the acquisition coefficients, filter state, and routing are restored. A gain-indexed electrically erasable programmable read-only memory (EEPROM) record stores the valid-stage mask, four DAC codes, version, and cyclic-redundancy check; a failed run may hold its bounded best codes but never overwrites a valid record.

B. Control Law and Calibration Parameters

For each stage, $\hat{y}_i[n]$ is the filtered signed ADC count proportional to $V_{\text{stage}} - V_{\text{ref}}$, r_i is the desired count, and $e_i[n]$ is the measured residual. Because the target is the reference level, $r_i = 0$ and $e_i[n] = \hat{y}_i[n] - r_i = \hat{y}_i[n]$. The residual is first mapped to the 8-bit DAC domain:

$$\begin{aligned} \varepsilon_i[n] &= \operatorname{round}\left(e_i[n]\frac{q_{\text{ADC}}}{q_{\text{DAC}}}\right) \\ &= \operatorname{round}\left(\frac{e_i[n]V_{\text{ADC,span}}N_{\text{DAC}}}{N_{\text{ADC}}V_{\text{DAC,span}}}\right), \end{aligned} \tag{2}$$

where $q_x = V_{x,\text{span}}/N_x$ is the step size and N_x is the number of quantization levels. All quantities used here are nominal datasheet values: $N_{\text{ADC}} = 2^{18}$, $N_{\text{DAC}} = 2^8$, $V_{\text{ADC,span}} = 5000$ mV, and $V_{\text{DAC,span}} = 4096$ mV as the ideal 256-bin quantization span, giving $q_{\text{ADC}} = 5000/2^{18}$ mV/count and $q_{\text{DAC}} = 16$ mV/code [7], [14].

The rounding in Eq. 2 gives an integer error in equivalent DAC codes. An overly narrow band can alternate indefinitely between adjacent codes when the ideal point is not realizable; the implementation therefore adopts the conservative one-step floor

TABLE I: Parameters used for the reported measurements. Deadbands are in equivalent DAC codes; S/L/R are settle/lock/refine samples at 2604 Hz, whereas Timeout is in seconds.

Stage	$\lvert K_i\rvert$	$\Delta_{i,\min}^{\text{impl}}$	Δ_i	$k_{\text{P},i}$	$k_{\text{I},i}$	S/L/R	Timeout (s)
PGA	1	1	18	10^{-3}	3×10^{-4}	512/512/1024	11.5
BP	1	1	3	10^{-4}	5×10^{-4}	512/1024/1024	17.3
SUM	7.89	8	10	10^{-4}	5×10^{-4}	512/512/1024	11.5
LP	6	6	8	10^{-4}	5×10^{-4}	1024/1024/2048	17.3

$$\Delta_{i,\min}^{\text{impl}} = \max(1, \lceil\lvert K_i\rvert\rceil), \quad (3) \qquad \bar{\varepsilon}_i[n] = \begin{cases} 0, & \lvert\varepsilon_i[n]\rvert \le \Delta_i, \\ \varepsilon_i[n], & \text{otherwise}, \end{cases} \quad (4)$$

Inside the band, Eq. 4 suppresses integration; lock and refinement still govern convergence. Table I lists adopted values satisfying $\Delta_i \ge \Delta_{i,\min}^{\text{impl}}$.

Under the assumptions of Section II, the architecture requires a measurable stage output and sufficient trim authority and range. The controller is written in positional proportional-integral (PI) form,

$$c_i^*[n] = c_{i,\text{base}} - s_i\left[\frac{k_{\text{P},i}}{\lvert K_i\rvert}\bar{\varepsilon}_i[n] + \frac{k_{\text{I},i}}{\lvert K_i\rvert}\sum_{q=0}^{n-1}\bar{\varepsilon}_i[q]\right], \tag{5}$$

where $c_{i,\text{base}}$ is the baseline feedforward code and $s_i = \operatorname{sgn}(K_i)$. The integral sum holds samples accepted before the current update and resets when the residual enters the deadband. The VDAC-to-node gain magnitudes $\lvert K_i\rvert$ used in Table I follow from each stage's topology and resistor ratios; at the two-op-amp instrumentation-amplifier PGA input [15], the reference reaches its measured node at unity DC gain ($K_{\text{PGA}} = 1$).

The command is clipped to that stage's configured limits within 0–255; at a limit, integration is suspended when the residual would push the command farther out, preventing windup. Convergence requires a fixed count of consecutive in-deadband samples, and an adjacent-code trial is kept only if it lowers the filtered residual. At timeout the best code is held and the stage flagged.

The proof-of-concept parameter set in Table I was selected empirically in preliminary bench tests, fixed across all reported conditions, and not systematically optimized; Section V-A discusses $\Delta_{\text{PGA}} = 18$. Finer trim can use a current-output DAC/resistor or buffered topology [14] without changing the sequence.

IV. Proof-of-Concept Evaluation

A. Initial Analog-Path Check

With no calibrated seismic source, the Ma-derived electronics were checked independently of the geophone. An Agilent 33220A waveform generator excited the front end in overlapping batches of 0.01 Hz–200 kHz logarithmic sinusoidal sweeps, while a Tektronix TDS 2004B recorded the stage inputs and outputs. The measured PGA-output-to-ADC-input analog response was defined by the LP-output/PGA-output ratio (CH4/CH1), so it covers the electronics downstream of the monitored PGA output.

TABLE II: Residual DC offset (%FS of the nominal 5-V ADC span), mean $\pm$ sample standard deviation over N=10 paired baseline-seeded restarts per reported board/gain cell. Within each cell, the upper and lower values correspond to the shared fixed-reference and calibrated conditions, respectively.

Stage	A, $\times1$	A, $\times4$	A, $\times16$	B, $\times16$	A, $\times32$	B, $\times32$	A, $\times50$	B, $\times50$
PGA	3.5 ± 0.2	2.4 ± 0.1	2.4 ± 0.1	2.1 ± 0.1	1.0 ± 0.2	1.0 ± 0.2	1.2 ± 0.3	2.5 ± 0.1
	3.4 ± 0.3	2.1 ± 0.3	1.9 ± 0.2	1.7 ± 0.1	0.5 ± 0.2	0.5 ± 0.2	1.0 ± 0.5	1.8 ± 0.3
BP	1.5 ± 2.2	4.9 ± 0.3	3.8 ± 0.9	0.66 ± 0.02	7.0 ± 0.4	0.66 ± 0.01	1.0 ± 0.7	0.65 ± 0.01
	1.2 ± 0.2	1.0 ± 0.2	1.5 ± 1.5	0.66 ± 0.02	1.0 ± 0.5	0.66 ± 0.01	0.9 ± 0.5	0.65 ± 0.02
SUM	$\mathbf{21.8 \pm 1.1}^{\dagger}$	6.7 ± 1.0	4.9 ± 1.5	4.5 ± 0.8	7.4 ± 0.3	8.0 ± 0.8	0.4 ± 0.2	0.9 ± 0.6
	0.8 ± 0.2	1.0 ± 0.3	1.4 ± 1.5	0.6 ± 0.5	0.6 ± 0.2	1.2 ± 0.3	$\mathbf{0.8 \pm 0.6}^{\ddagger}$	0.9 ± 0.6
LP	$\mathbf{45.8 \pm 3.1}^{\dagger}$	$\mathbf{42.6 \pm 4.2}^{\dagger}$	$\mathbf{29.6 \pm 6.8}^{\dagger}$	3.0 ± 0.1	$\mathbf{12.4 \pm 1.3}^{\dagger}$	2.9 ± 0.1	4.1 ± 2.1	2.9 ± 0.1
	3.2 ± 0.9	3.9 ± 0.8	2.3 ± 1.4	0.7 ± 0.1	1.7 ± 0.7	0.4 ± 0.2	1.2 ± 0.4	0.7 ± 0.2

Lower is better. †Fixed reference $\geq$ 10%FS. ‡The marked calibrated mean is larger than its baseline; equality after rounding is not marked (Section V-A).

For the approximate sensor-to-ADC shape in Fig. 2, the measured electronics response was cascaded with the nominal SM-24 acceleration model [6]. All magnitude curves are normalized: the measured composite is set to 0 dB at its maximum; the ideal chain and 0–2 kΩ tuning envelope use that same reference, whereas the standalone geophone acceleration and velocity curves are each normalized to their own maximum. Local coherence, computed from the squared normalized cross-term of the analytic input and output within logarithmic sweep bins, first falls below 0.9 at approximately 0.84 kHz; the dash-dotted vertical line marks the rounded 1-kHz onset. Above this point, strong attenuation and nonmonotonic coherence prevent a reliable amplitude comparison with the model. The growing disagreement is therefore treated as measurement-limited rather than a verified change in circuit behavior. This is an initial circuit-level consistency check, neither a calibrated sensor-chain characterization nor validation of the offset-calibration method. End-to-end measurements with a calibrated seismic source remain future work.

B. Prototype and Comparison Conditions

The main comparison contains eight board–gain cells, each measured through ten paired restarts: all five PGA settings on Board A and the $\times16$, $\times32$, and $\times50$ settings on Board B. Thus, N=10 represents repeated restarts of the same board and gain, not ten independent devices. The shared fixed-reference baseline, $\boldsymbol{c}_{\mathrm{base}}$, was selected empirically on Board B at $\times50$ and then used unchanged on Board A and at the other gains. For each calibrated condition, the four stages were processed sequentially and the resulting DAC codes were frozen during acquisition. Post-calibration EEPROM writes were disabled in these measurements, so every restart began again from $\boldsymbol{c}_{\mathrm{base}}$.

Board B at $\times1$ and $\times4$ were omitted from the eight-cell summary: their baseline signals fell within the normal deadbands, so the algorithm applied no corrective updates and no distinct after-calibration condition exists. A separate narrower-deadband diagnostic (Section V-A) confirmed the expected response.

C. Residual DC-Offset Measurement and Results

The LP output is the primary outcome, delivering the conditioned signal to the acquisition ADC; PGA, BP, and SUM document sequential correction and expose near-rail intermediate operating points.

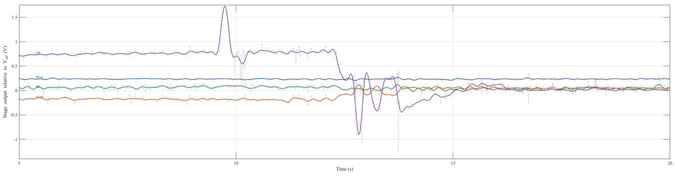

Fig. 4: Stage deviations relative to V_{ref} during calibration at $\times16$ (total differential gain $\times32$). Dashed light-gray traces are oscilloscope measurements; colored curves are resampled 128-tap-FIR reconstructions, not internal logs (Section V-A).

At each setting, a four-channel Tektronix TDS 2004B (8-bit, 60 MHz) recorded V_{mean} at PGA, BP, SUM, and LP over a 1s window after the transient settled. All probe grounds were tied to $V_{\mathrm{ref}} = V_{\mathrm{DDA}}/2$, so each channel measured $V_{\mathrm{stage}} - V_{\mathrm{ref}}$; because the USB host was unearthed while the oscilloscope ground was at protective earth, this also fixed V_{ref} to earth potential [16]. USB-supplied V_{DDA} ran slightly below 5 V, and other leakage paths were not characterized. Within each restart the fixed-reference condition was recorded first and the calibrated condition second, preserving the pairing; as this order was never reversed, slow drift cannot be fully separated from the calibration effect.

Table II reports the mean and sample standard deviation of $100|V_{\mathrm{mean}}|/5\,\mathrm{V}$ over ten paired restarts. Values near a supply rail may be clipped and indicate saturation rather than exact linear offsets. These descriptive statistics characterize within-configuration repeatability only: no oscilloscope uncertainty budget was performed, and displayed digits do not imply sub-millivolt accuracy.

Figure 4 applies the 128-tap calibration FIR to oscilloscope records resampled at the controller rate; the colored curves approximate the controller's filtered outputs but are external reconstructions, not logs, excluding native sampling, switching, resets, and fixed-point arithmetic. Table II uses later oscilloscope records, not the controller's final ADC samples.

Across the eight reported board$\times$gain cells, the calibrated LP sample mean is 0.4–3.9%FS versus 2.9–45.8%FS under the shared fixed-reference baseline; individual calibrated observations span 0.1–5.2%FS (Table II). The LP mean was lower

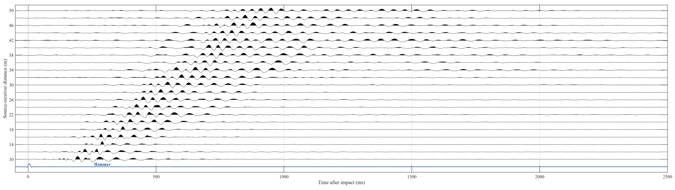

Fig. 5: Wiggle-trace field record acquired after foreground calibration; the trigger and first-arrival moveout demonstrate multichannel operation over the 10–50 m line.

after calibration in every reported cell, without board- or gain-specific controller retuning; a one-sided sign test on the paired LP magnitudes rejects the no-reduction null in all eight cells (10 of 10 restarts lower per cell, exact $p \approx 10^{-3}$).

D. Field Operation Demonstration

This field trial was an operational demonstration, not a controlled offset or surface-wave comparison. Fig. 5 shows a 10–50 m record with each geophone planted and kept stationary during the preceding calibration; the resolved first-arrival moveout demonstrates operation. Across laboratory and field measurements, the only condition that triggered a timeout was powering a board while its geophone was moving, whose AC transient saturated the chain and violated the quasi-static assumption.

V. Discussion

Unlike precision front ends characterized near the sensor's Johnson-noise limit [17], this method trades component precision for measurement and digital assistance, and unlike software subtraction it physically recenters observable additive DC errors within DAC authority and range. The comparison uses one shared fixed-reference baseline rather than optimized per-condition codes and evaluates the complete sequence, not each trim; per-stage ablation remains future work.

Continuous DC-servo loops introduce a settling-versus-bandwidth tradeoff [18]; the foreground controller instead holds constant DAC outputs during acquisition and avoids a continuously active feedback path.

A. Non-Optimized PGA Acceptance

PGA offset is only modestly smaller than baseline: with $|K_{\mathrm{PGA}}| = 1$, the non-optimized $\Delta_{\mathrm{PGA}} = 18$ accepts integer-error buckets through ± 288 mV (a continuous edge near ± 296 mV after rounding). The 0.24-V level in Fig. 4 lies inside it, consistent with stop-on-band rather than instability, but sets no accuracy floor. A separate narrower-deadband diagnostic on Board B at $\times 4$ (three restarts, LP mean $2.98 \rightarrow 0.84\%$FS), run outside the main comparison and not reported in Table II, indicates that tighter bands reduce the LP residual, though its different bands and small sample preclude comparison with the reported cells. The only increase at the reported precision, Board A SUM at $\times 50$ (0.4 to 0.8%FS), reflects band acceptance and does not affect the downstream LP result.

B. Timing and Scope

Visible correction in Fig. 4 spans about 7 s; an unlogged bench observation placed completion near 10 s, but runs were not systematically timed. The four timeout windows sum to 57.6 s, and adding the listed settling and refinement gives about 60.5 s, excluding stored-code verification and switching overhead. No laboratory timeout was observed.

This exploratory two-board study cannot establish population scaling or statistical significance, and fixed ordering leaves drift as a confound. Future work requires more independent boards, randomized order, and confirmatory statistical tests; per-stage ablation and parameter-sensitivity studies; characterization of finite DAC resolution, noise and distortion, sustained vibration, thermal drift, and slow filter-state transients; controlled before/after field comparisons; and evaluation of the calibration subsystem within the complete distributed IoT platform.

VI. Conclusion

This exploratory two-board proof of concept shows that sequential four-VDAC foreground calibration, with codes frozen during acquisition, substantially lowers the residual DC offsets across the conditioning stages—most consequentially at the LP output delivered to the acquisition ADC—relative to a shared fixed-reference baseline (Section IV). The evidence is descriptive feasibility, not population-level validation; the 10–50 m field record confirms integration, not comparative performance. The studies required to establish generality are outlined in Section V.

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